Investigating Low PMTCT Service Uptake Among Pregnant Women in Nigerian Communities

Obiageli Iweanoge

Women Ascension Learning and Understanding AI and STEM Department

Supervisor: Dr. Maheen Amjad

HoD: Amel Barkat

AHoD: Nargis Ahmadi

Submitted in partial fulfilment of the requirements for the fellowship of
WALUAIS Programme

Centre for Learning Innovation
Women Ascension
October, 2026

Keywords

Artificial Intelligence, data analytics, digital literacy, STEAM thinking, technology, maternal health, community impact

Abstract

This project documents my individual investigation into low PMTCT service uptake among pregnant women in communities across Nigeria as part of the WALUAIS Batch III fellowship. Although PMTCT services are available at healthcare facilities, some pregnant women do not access HIV testing and treatment during pregnancy, increasing the risk of mother-to-child transmission of HIV.

I chose this problem because my previous analysis of PMTCT data showed differences in service uptake across communities. In Individual Assignment 1, I used systems thinking to examine the problem and identified stigma, trust, transportation, and limited use of community data as possible factors. These were treated as assumptions that still needed evidence.

In Individual Assignment 2, I selected transportation as one area to investigate. I used a Google Form exercise with 8 sample responses and collected information on travel time, transport cost, and missed or delayed visits. I then used AI to help identify patterns and checked the results myself. The evidence suggested that transport difficulties may affect access, but the small sample was not enough to confirm this as a major barrier.

This investigation showed me the importance of testing possible causes before drawing conclusions. It also provides a starting point for further investigation as the project develops.

Table of Contents

Keywords 2
Abstract 3
Table of Contents 4
List of Abbreviations 6
Statement of Original Authorship 7
Acknowledgements 8
Chapter 1: Introduction 9
1.1 Background 9
1.2 Context 9
1.3 Purposes 9
1.4 Significance, Scope & Definitions 10
1.5 Project Outline 10
Chapter 2: Programmes Review 12
2.1 Historical Background 12
2.2 Description of Programmes 12
2.2.1 Individual Assignment 1: Problem Discovery & Systems Thinking 12
2.2.2 Individual Assignment 2: Data Thinking & Evidence 12
2.3 Organizing & Implementation 13
2.4 Learning Outcomes 14
2.5 Summary & Implications 14
Chapter 3: Visual Representation of Programmes 16
3.1 Individual Assignment 1: Problem Discovery & Systems Thinking 16
3.2 Individual Assignment 2: Data Thinking & Evidence 18
3.3 Team Project: Problem Definition 23
Chapter 4: Programme Management & Coordination 24
4.1 Overview of Remote Programmes 24
4.2 Planning & Preparation 24
4.3 Technology & Platforms Used 24
4.4 Team Roles & Responsibilities 25
4.5 Participant Engagement & Interaction 25
4.6 Challenges in Remote Implementation 25
4.7 Outcomes & Feedback 25
4.8 Lessons Learned & Recommendations 26
Chapter 5: Analysis 27
5.1 Summary of Findings 27
5.2 Implications & Insights 27
5.3 Recommendations for the Future 27
Chapter 6: Conclusion 28
Bibliography /References 29
Weekly Progress Tracker 30
Appendices 36
Appendix A – Attendance and Participation Records 36
Appendix B – Feedback Forms 38
Appendix C – Raw Data & Notes 41

List of Abbreviations

PMTCT - Prevention of Mother-to-Child Transmission of HIV

AI - Artificial Intelligence

WALUAIS - Women Ascension Learning and Understanding AI and STEM Department

HIV – Human Immunodeficiency Virus

STEAM – Science, Technology, Engineering, Arts and Mathematics

STEM – Science, Technology, Engineering and Mathematics

Statement of Original Authorship

The work contained in this project has not been previously submitted to meet requirements for an award at this or any other higher education institution. To the best of my knowledge and belief, the project contains no material previously published or written by another person except where due reference is made.

Signature: _____Obiageli Iweanoge_____

Date: _____October 7, 2026_____

Acknowledgements

I sincerely appreciate everyone who has supported me throughout my WALUAIS Batch III fellowship journey.

I am grateful to my Supervisor, Dr. Maheen Amjad, Founder of Women Ascension, for creating this opportunity and providing a platform for women to learn, grow, and explore new areas of technology.

I also appreciate Amel Barkat, Head of Department (WALUAIS), for her leadership and guidance throughout the programme.

My sincere thanks to Nargis Ahmadi, Assistant Head of Department and Team Coordinator, for her coordination, support, and encouragement.

I am also grateful to Grace Nakiyemba for facilitating the STEAM Problem-to-Project training series and guiding me through the frameworks I used in developing this project, and to my fellow WALUAIS participants for the learning, discussions, and support we have shared throughout the fellowship.

This experience has given me the opportunity to learn, ask questions, examine problems more carefully, and continue building my skills.

Chapter 1: Introduction

1.1 Background

Prevention of Mother-to-Child Transmission (PMTCT) services are available to help pregnant women living with HIV access interventions that reduce the risk of transmission to their babies during pregnancy, delivery and breastfeeding ***(World Health Organization, n.d.)***.

When testing and treatment are missed, the risk of vertical transmission increases, and opportunities for early intervention may be lost.

This problem is not limited to one facility or community, as PMTCT service uptake varies across communities.

1.2 Context

I chose this problem because an earlier PMTCT data analysis I carried out showed differences in service uptake across communities. That analysis showed me what was happening, but not why.

Through my WALUAIS individual project, I had the opportunity to investigate the problem further using the STEAM Problem-to-Project frameworks. I used systems thinking and root cause analysis to examine possible factors and then used evidence to test one of them. This approach helped me move from observing a pattern in the data to asking what might be contributing to it.

The project connects with Women Ascension's focus on helping women develop practical skills in AI, data, technology, and STEM. It also aligns with the Sustainable Development Goal on Good Health and Well-being (SDG 3), particularly through its focus on maternal health and access to health services. This connects with my department's focus on applying AI, data, and technology to health and community issues.

1.3 Purposes

The purpose of this project is to investigate the problem of low PMTCT service uptake before attempting to design a solution.

Specifically, the project aims to:

- Define the problem clearly and identify who is affected and where it occurs.
- Examine possible factors contributing to low uptake using systems thinking.
- Separate what is supported by evidence from what is still an assumption.
- Select one assumption for testing and gather small-scale evidence.
- Look for patterns in the evidence, using AI only as an analysis assistant, and check its output myself.
- Reflect on what the evidence tells me about the problem and whether it changes my initial understanding.

This project does not aim to propose or implement a solution at this stage.

1.4 Significance, Scope & Definitions

Significance

This problem matters because early HIV testing and treatment during pregnancy can help reduce the risk of vertical transmission ***(World Health Organization, 2025)*****.** When women do not access these services, opportunities for prevention and early care may be missed.

The investigation may be useful to pregnant women, healthcare workers, and people involved in planning PMTCT programmes because understanding the factors affecting service uptake is an important first step before deciding what should be done.

Scope

The scope of this project is limited to my individual investigation of low PMTCT service uptake among pregnant women. In Assignment 1, I examined possible factors that could contribute to the problem. In Assignment 2, I selected transportation for further investigation using a small Google Form survey with 8 sample responses created for the exercise.

Other factors identified in Assignment 1, including stigma and trust, remained possible factors but were not tested in the evidence gathered for Assignment 2.

Key terms

- **PMTCT (Prevention of Mother-to-Child Transmission):** Health services and interventions that help prevent HIV transmission from an HIV-positive mother to her child during pregnancy, delivery, or breastfeeding.
- **Service uptake:** The extent to which pregnant women use available PMTCT services, including HIV testing and treatment.
- **Vertical transmission:** The transmission of HIV from mother to child.

1.5 Project Outline

This project developed from an individual investigation into a team project through the WALUAIS STEAM Problem-to-Project activities.

In Assignment 1: Problem Discovery & Systems Thinking, I defined the PMTCT problem, examined it using a systems approach, and looked at factors such as people, processes, resources, technology, environment, policies, access, and behaviour. I also separated what I knew from what I was assuming. This led to three questions for further investigation, including one about transportation.

In Assignment 2: Data Thinking & Evidence, I selected the transportation question for further investigation. I collected small-scale evidence using a Google Form survey, looked at travel time,

transport cost, and missed or delayed visits, and used AI to help identify patterns. I then checked the AI output myself and developed three insights and a simple visual from the exercise.

Chapter 2 will describe these activities and what I found. Chapter 3 will present the related visual evidence.

Chapter 2: Programmes Review

2.1 Historical Background

This individual project began with an earlier analysis of PMTCT data, which showed differences in service uptake across communities. The analysis helped me see a problem, but it did not explain what was contributing to those differences.

The STEAM Problem-to-Project training gave me a way to investigate the problem further. In Individual Assignment 1, I used problem discovery and systems thinking to examine possible factors. In Individual Assignment 2, I moved from possible explanations to a small evidence-gathering exercise.

This made me pay more attention to what I knew, what I was assuming, and what I still needed to find out.

2.2 Description of Programmes

2.2.1 Individual Assignment 1: Problem Discovery & Systems Thinking

The first assignment focused on understanding the PMTCT service uptake problem before thinking about a solution.

I defined the problem and examined it using systems thinking. I considered different parts of the system, including people, processes, resources, technology, environment, policies, access, and behaviour.

The exercise identified several possible factors that could affect PMTCT service uptake, including stigma, trust, transportation, and limited use of community data. These were treated as possible factors rather than confirmed causes.

I then developed three questions for further investigation:

1. What role do stigma and trust in healthcare providers play in women's decisions to access PMTCT services?
2. How do transportation challenges affect pregnant women's access to PMTCT services?
3. Can community-level health data help identify which specific areas or groups have the lowest uptake, and why?

For Assignment 2, I selected the question relating to transportation.

2.2.2 Individual Assignment 2: Data Thinking & Evidence

The second assignment focused on using evidence to examine the transportation question.

I used a Google Form with 8 sample responses for the exercise. The questions covered travel time to the health facility, transportation cost, difficulty attending antenatal or PMTCT visits, missed or delayed visits, and the main transportation challenge reported.

The responses showed that:

- **5 of 8 sample responses** said transportation challenges had made it difficult to attend an antenatal or PMTCT visit.
- **4 of 8 sample responses** reported that they had missed or delayed a visit because of transportation problems.
- **4 of 8 sample responses** identified high transport cost as their main transportation challenge.
- **3 of 8 sample responses** reported travel times of 31–60 minutes.
- **3 of 8 sample responses** reported spending ₦1,001–₦2,000 on transportation.
- **2 of 8 sample responses** reported travelling for more than one hour.
- **2 of 8 sample responses** reported spending more than ₦2,000 on transportation.

These results suggest that transportation was a concern within this sample. However, the sample was small and was created for the exercise, so the findings cannot be used to conclude that transportation is a major barrier for pregnant women generally.

2.2.3 Team Project: Problem Definition

Following the individual assignments, the project progressed to the team stage. The team reviewed the PMTCT issue and selected *low PMTCT service uptake among pregnant women* as the problem to work on.

I worked with one other team member to discuss and finalize the team problem statement, defining the people affected, the setting, the problem, and why it matters.

This marked the transition from the individual investigation to collaborative project development. The team then began considering possible solutions based on the defined problem.

2.3 Organizing & Implementation

The individual investigation followed the two assignments in sequence.

In Assignment 1, I first examined the problem and identified possible factors instead of immediately choosing one as the cause. I then developed questions that could be investigated further.

In Assignment 2, I selected transportation as the factor to investigate. I used a Google Form and sample data to collect information related to travel time, transport cost, and missed or delayed visits.

I reviewed the evidence and then used AI to help identify patterns. I checked the AI output against the actual data rather than accepting the results without verification.

The project then progressed to the team stage. I worked with one other team member to discuss and finalize the team problem statement. The team then began considering possible solutions based on the defined problem.

My process was to identify the problem, consider possible factors, ask a question, gather evidence, look for patterns, and then move into the team project.

2.4 Learning Outcomes

Through the two individual assignments, I learned to look at a problem before looking for a solution. I learned that a possible cause is not the same as a confirmed cause. For example, stigma and trust appeared as possible factors in Assignment 1, but I did not have evidence from Assignment 2 to confirm either one.

I also learned how to take a broad problem and reduce it to one question that could be investigated with a small amount of data. The transportation exercise helped me practise looking at data through simple measures such as travel time, transport cost, and missed or delayed visits. I also learned that AI can assist with identifying patterns, but its output still needs to be checked against the original data.

During the team stage, I learned the importance of discussing and refining a problem statement before moving further into solution development. Working with another team member helped me apply the problem-definition approach from the individual stage to a team project.

The main lesson for me was to look for evidence before reaching a conclusion.

2.5 Summary & Implications

The individual investigation provided an initial look at transportation as a possible factor affecting PMTCT service access.

Within the 8 sample responses, 5 sample responses said transportation challenges had made it difficult to attend an antenatal or PMTCT visit, while 4 sample responses reported missing or delaying a visit because of transportation problems. High transport cost was also the most commonly reported main transportation challenge, identified by 4 sample responses.

However, these results do not establish transportation as a major barrier among pregnant women generally. The data was based on only 8 sample responses created for the exercise.

The main implication of the individual investigation was therefore not that a cause had been identified, but that transportation remained an area that could require further investigation. The project then progressed to the team stage, where the team selected low PMTCT service uptake among pregnant women as the problem for further project development.

Chapter 3: Visual Representation of Programmes

This chapter presents visual and documentary evidence from my individual WALUAIS learning activities. The activities documented here are from the individual stage of the fellowship.

3.1 Individual Assignment 1: Problem Discovery & Systems Thinking

Session date: 4 August 2026 · Submitted: 9 August 2026

Platform: Fellow Growth Evidence Tracker

Audience: Individual submission, accessible to WALUAIS HoD / Team coordinator with editor rights via the Fellow Growth Evidence Tracker

Participants: Individual assignment

Evidence Documents | Link

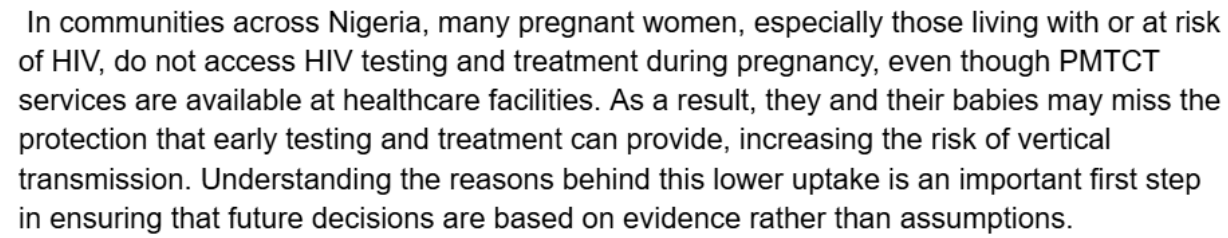


Figure 1: Individual problem statement on low PMTCT service uptake among pregnant women.

System Area	Possible Factors to Investigate
People & Stakeholders	Pregnant women, healthcare workers, CHEW's,families,TBA's and religious leaders.
Processes	Referral processes, follow-up care, continuity of PMTCT services, patient flow
Resources	HIV test kits, medication, trained healthcare workers, funding
Technology & Data	Data collection, reporting systems, use of community health data to identify underserved areas
Environment	Community location, rural or urban setting, road conditions, security, local infrastructure
Policies & Institutions	Implementation of PMTCT programmes, health system support, coordination between stakeholders
Access	Availability of services, clinic hours, ease of reaching healthcare facilities
Behaviours & Community Factors	Awareness of PMTCT services, trust in healthcare providers, stigma, family or community support

Figure 2: Systems thinking analysis of factors that may affect PMTCT service uptake.

1. What role do stigma and trust in healthcare providers play in women's decisions to access PMTCT services?

2. How do transportation challenges affect pregnant women's access to PMTCT services?

3. Can community-level health data help identify which specific areas or groups have the lowest uptake, and why?

Figure 3: Investigation questions developed from the individual PMTCT problem.

3.2 Individual Assignment 2: Data Thinking & Evidence

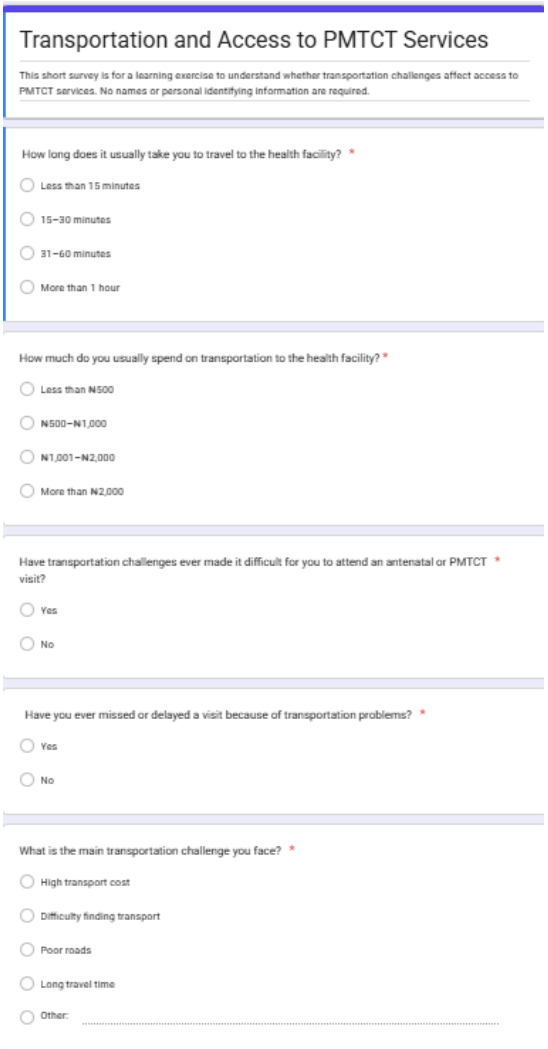
Session date: 11 August 2026 · Submitted: 13 August 2026

Platform: Fellow Growth Evidence Tracker

Audience: Individual submission, accessible to **WALUAIS** HoD / Team coordinator with editor rights via the Fellow Growth Evidence Tracker

Participants: Individual assignment

Evidence Documents | Link



Transportation and Access to PMTCT Services

This short survey is for a learning exercise to understand whether transportation challenges affect access to PMTCT services. No names or personal identifying information are required.

How long does it usually take you to travel to the health facility? *

- Less than 15 minutes
- 15–30 minutes
- 31–60 minutes
- More than 1 hour

How much do you usually spend on transportation to the health facility? *

- Less than ₦500
- ₦500–₦1,000
- ₦1,001–₦2,000
- More than ₦2,000

Have transportation challenges ever made it difficult for you to attend an antenatal or PMTCT visit? *

- Yes
- No

Have you ever missed or delayed a visit because of transportation problems? *

- Yes
- No

What is the main transportation challenge you face? *

- High transport cost
- Difficulty finding transport
- Poor roads
- Long travel time
- Other: .....

Figure 4: Google Form used for the transportation evidence exercise.

Form_Responses

	Timestamp	How long does it usually take you to travel to the health facility?	How much do you usually spend on transportation to the health facility?	Have transportation challenges ever made it difficult for you to attend an antenatal or PMTCT visit?	Have you ever missed or delayed a visit because of transportation problem	What is the main transportation challenge you face?
2	8/13/2026 17:27:21	15–30 minutes	₦500–₦1,000	No	No	High transport cost
3	8/13/2026 17:28:23	31–60 minutes	₦1,001–₦2,000	Yes	Yes	High transport cost
4	8/13/2026 17:28:58	Less than 15 minutes	Less than ₦500	No	No	Difficulty finding transport
5	8/13/2026 17:29:50	31–60 minutes	₦1,001–₦2,000	Yes	Yes	Poor roads
6	8/13/2026 17:30:15	More than 1 hour	More than ₦2,000	Yes	Yes	High transport cost
7	8/13/2026 17:30:48	15–30 minutes	₦500–₦1,000	No	No	Long travel time
8	8/13/2026 17:31:09	More than 1 hour	More than ₦2,000	Yes	Yes	High transport cost
9	8/13/2026 17:31:37	31–60 minutes	₦1,001–₦2,000	Yes	No	Poor roads

Figure 5: Sample data showing travel time, transport cost, and reported difficulties attending visits.

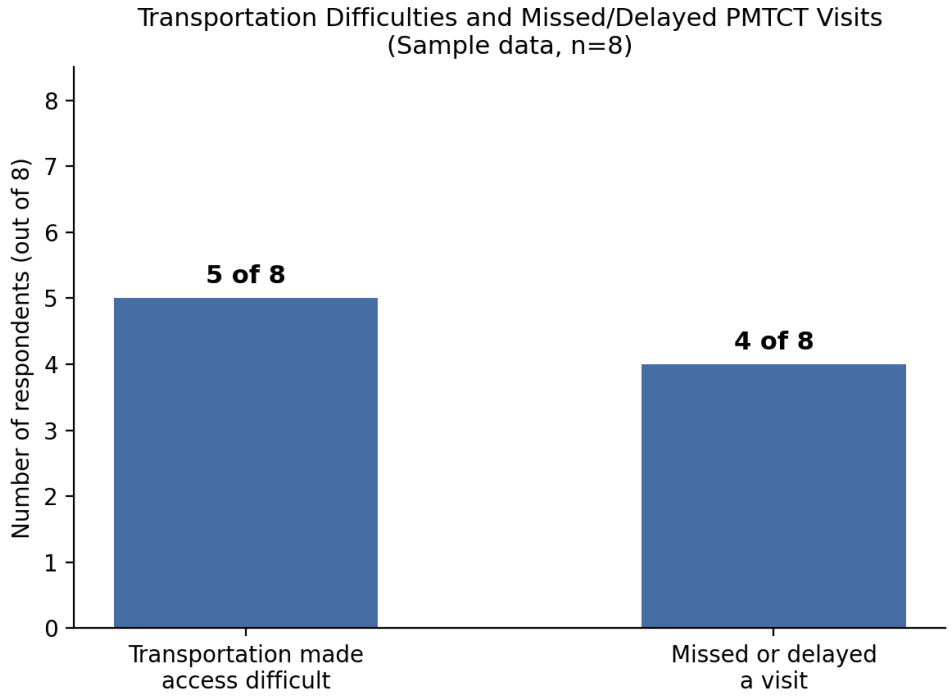


Figure 6: Transportation difficulties and missed or delayed PMTCT visits, based on 8 sample responses.

5 of 8 sample responses reported that transportation made it difficult to attend antenatal or PMTCT visits, and 4 of 8 sample responses reported missing or delaying a visit because of transportation problems. Sample data, not collected from real respondents.

INSIGHT 1

What does the evidence show?

5 of 8 sample responses said transportation made it difficult to attend
antenatal or PMTCT visits. In this small data sample, transportation
appears to be a possible barrier to accessing care.

Why does this matter for the problem?

It suggests that transportation may be affecting access to PMTCT services
and should be investigated further.

INSIGHT 2

What does the evidence show?

4 of 8 sample responses reported missing or delaying a visit because of
transportation problems.

Why does this matter for the problem?

It suggests that transportation difficulties may affect whether women attend
their scheduled visits.

INSIGHT 3

What does the evidence show?

5 of 8 sample responses who reported transportation difficulties had travel
times of 31 minutes or more.

Why does this matter for the problem?

it suggests that longer travel times may be linked to transportation difficulties,
but the small data sample is not enough to confirm this.

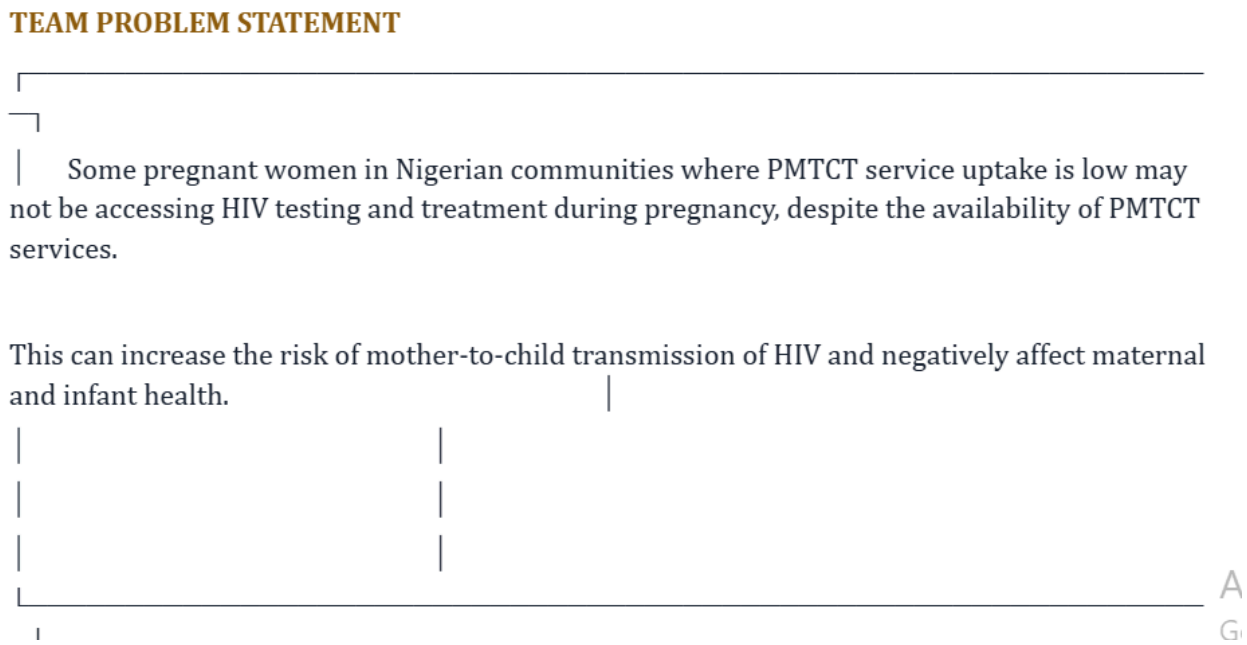
Figure 7: Three insights developed from the transportation evidence.

3.3 Team Project: Problem Definition

Session/Project stage: Team Project
Activity: Team Problem Statement
Participants: Obiageli Iweanoge and Sophia Ugwuanya

> The team reviewed the PMTCT problem and developed a problem statement focused on pregnant women in Nigerian communities where PMTCT service uptake is low. The finalized statement identifies the possible gap between the availability of PMTCT services and access to HIV testing and treatment during pregnancy, and explains why this matters for maternal and infant health.

Figure 8: Finalized team problem statement on low PMTCT service uptake among pregnant women.



Chapter 4: Programme Management & Coordination

4.1 Overview of Remote Programmes

The WALUAIS Batch III fellowship was conducted remotely, allowing participants to learn and complete activities using online platforms. The remote format supported participation in training sessions, individual assignments, documentation, and interaction with other fellows.

My individual project activities were completed as part of the STEAM Problem-to-Project training. The work progressed from problem discovery and systems thinking to data and evidence gathering, responsible AI considerations, and project blueprint development.

4.2 Planning & Preparation

I prepared for the fellowship by setting up my learning environment, organising my Google Drive, accessing the learning platforms, and maintaining my Fellow Growth Evidence Tracker.

For the individual project, I followed the sequence provided in the training:

Problem → Possible factors → Question → Evidence → Pattern → Insight

This helped me move from identifying the PMTCT problem to investigating one possible factor, transportation.

4.3 Technology & Platforms Used

The fellowship activities involved several digital tools and platforms. These included:

- Zoom for remote training sessions
- WhatsApp for communication and coordination
- Google Drive for organising project materials and evidence
- Google Forms for the transportation evidence exercise
- AI tools to support analysis and identify patterns
- Fellow Growth Evidence Tracker for documenting learning activities and submissions

For the transportation exercise, I used Google Forms to gather the sample responses and then reviewed the data before using AI to assist with identifying patterns.

4.4 Team Roles & Responsibilities

During the individual stage, I was responsible for developing and documenting my own project activities. This included defining the PMTCT problem, carrying out the systems thinking exercise, developing investigation questions, collecting and reviewing the sample evidence, analysing the findings, and documenting the learning.

The team project phase later introduced shared responsibilities such as research and evidence, user needs, AI and tools, prototype development, testing, documentation, and presentation. The WALUAIS guide states that responsibilities should be linked to specific deliverables and that every Fellow should contribute.

4.5 Participant Engagement & Interaction

My participation included attending the scheduled training sessions, completing individual assignments, documenting my learning, and engaging with the learning materials.

The records in Appendix A document my participation in the STEAM Problem-to-Project sessions and subsequent learning sessions held between August and October 2026.

The feedback records also document learning from these sessions, including problem definition, evidence gathering, responsible AI, and moving from a problem towards a project blueprint.

4.6 Challenges in Remote Implementation

One challenge was determining whether the evidence collected was sufficient to support a conclusion. The transportation exercise used only 8 sample responses, which meant that the findings could not be generalised to pregnant women generally.

Another challenge was recognising that some possible barriers may not be visible in available data. During the responsible AI session, I reflected on issues such as privacy, missing voices, and invisible barriers. This reinforced the need for human judgement when using AI and data to understand community problems.

4.7 Outcomes & Feedback

The individual project taught me to investigate the problem with evidence before thinking about a solution.

The transportation exercise showed that transportation was a concern within the small sample. However, it did not establish transportation as the main cause of low PMTCT uptake. This helped me understand the difference between a possible factor and a confirmed cause.

4.8 Lessons Learned & Recommendations

The project taught me to begin with the problem rather than immediately selecting a technology or solution. I also learned to distinguish assumptions from evidence and to verify AI-generated analysis against the original data.

Future project work should allow enough time to gather and test evidence before developing a solution.

Chapter 5: Analysis

5.1 Summary of Findings

The individual investigation showed that low PMTCT service uptake is a problem that requires further investigation before a specific cause or solution can be identified.

The initial PMTCT data analysis showed differences in service uptake across communities. The systems thinking exercise identified several possible factors, including stigma, trust, transportation, and limited use of community data. These were treated as possible factors rather than confirmed causes

The transportation investigation provided some evidence that transportation may affect access. In the 8 sample responses, 5 reported that transportation had made it difficult to attend an antenatal or PMTCT visit, while 4 reported missing or delaying a visit because of transportation problems. However, the data was created for the exercise and was too small to establish transportation as a major barrier generally.

5.2 Implications & Insights

My main lesson was that I should not treat an assumption as a fact without evidence.

The investigation helped me move from observing differences in PMTCT uptake to asking specific questions about what might contribute to the problem. It also showed me that AI can support data analysis, but its output needs to be checked against the original data.

The findings also show that transportation should not yet be treated as the main cause of low PMTCT uptake. Instead, it is one possible factor that may require further investigation.

5.3 Recommendations for the Future

Future investigation should use a larger and more appropriate sample to examine transportation and other possible barriers to PMTCT service uptake.

Further research should also consider factors identified during the systems thinking exercise, including stigma, trust, access, community support, and the use of community-level data.

Any future solution should be based on stronger evidence and tested with the people it is intended to support.

The use of AI should continue to include human review, verification of important outputs, protection of personal information, and careful consideration of barriers that may not appear in the available data

Chapter 6: Conclusion

This project helped me investigate low PMTCT service uptake among pregnant women using a step-by-step, evidence-based approach.

I began by examining the problem through systems thinking and identifying possible factors that could affect access to PMTCT services. I then selected transportation as one factor to investigate and used a small evidence-gathering exercise to examine travel time, transport cost, and missed or delayed visits.

The findings suggested that transportation may affect access within the sample, but the evidence was not sufficient to confirm transportation as a major barrier among pregnant women generally. This was an important lesson in the need to test assumptions before drawing conclusions

The project also strengthened my understanding of responsible AI use. I learned that AI can assist with analysing information and identifying patterns, but human judgement and verification remain necessary.

By the end of the project, I was more comfortable moving from a broad problem to a specific question, checking evidence, and stating what I still needed to investigate. This gives me a clearer starting point for future work on PMTCT service access and community-level barriers.

Bibliography /References

Iweanoge, O. (2026). What the Number Doesn't Tell You:

Final Chapter of My PMTCT Journey. LinkedIn.

World Health Organization. (2025). *Strengthening the delivery of HIV testing services and linkage to care during pregnancy and postpartum for elimination of mother-to-child transmission*. **World Health Organization.**

World Health Organization. (n.d.). *Mother-to-child transmission of HIV*. **World Health Organization**.

Weekly Progress Tracker

Week	Activities Planned	Activities Completed	Challenges Faced	Solutions / Adjustments	Proof (pictures/links /screenshots)	Supervisor's Notes
Week 1	Understand the fellowship structure and learning expectations.	Navigated the learning platforms, explored basic AI concepts, updated my LinkedIn profile, and organized my Google Drive for the fellowship..	None Recorded.	Completed platform setup and began using the Growth Tracker.	https://www.linkedin.com/in/obiageli-iweanoge	
Week 2	Set up / share my learning environment and Growth Tracker.	Set up and organized the Growth Evidence Tracker and Google Drive for documenting my learning.	None Recorded.	More practice examples focused to strengthen my understanding of AI fundamentals.		
Week 3	Build a solid foundation in AI concepts.	Updated my LinkedIn profile, completed the quiz, and participated	None Recorded.	Applied AI concepts while preparing for STEAM activities.	https://www.linkedin.com/in/obiageli-iweanoge	

		in the learning activities.				
Week 4	Learn how to identify and define real problems before proposing solutions.	Applying the 5 Whys technique Mapping stakeholders Distinguishing assumptions from evidence Drafting a problem statement	Realized that some initial assumptions were not yet supported by evidence, a reminder to validate ideas before drawing conclusions.	Sought feedback on the problem statement, guidance on validating assumptions with evidence, and clarification on stakeholder mapping. Next goal: refine the problem statement using evidence and stakeholder insights.		
Week 5	Understand how to move from assumptions to evidence using data.	Created a Google Survey Form, reviewed sample data, identified patterns, and used AI to support the analysis.	I found it difficult to know what the data could actually prove, especially with a small sample.	More practice interpreting data and guidance on knowing when evidence is enough. Learned not to treat an assumption as a fact without evidence.		

Week 6	Understand how to apply responsible AI principles when thinking about solutions.	Applied the responsible AI questions to the PMTCT problem, especially around privacy, invisible barriers, risks, and human oversight.	Difficult to think about barriers that may not be captured in available data due to privacy limitations	More practical examples of applying responsible AI principles to real projects. Key realization: some barriers may remain invisible in the data, so an AI solution may not fully address the problem.	Responsible AI session notes and 10 Rule Ethical AI Checklist. 	
Week 7	Learn how to turn a problem, evidence and safeguards into a clear and feasible project blueprint.	Applying the blueprint concepts to the PMTCT team project.	Applied the project blueprint to define the team project direction and next steps.	Guidance and hands-on practice building the prototype. Key takeaway: design the human experience first, then choose the technology		
Week 8	AI Tools Lab: Discover Your Stack	Screenshots showing AI tool use for project planning, verification, and prototype development..	Understanding which AI tool was most suitable for different tasks.	Used each tool for a specific purpose and compared the usefulness of the outputs		

Week 9	AI Automation & Workflows in Practice	Practised building an automation workflow using Make.com to monitor a shared document and send notifications.	Understanding permissions and setting up the trigger correctly.	Tested the workflow step by step and adjusted the setup until the notification worked.		
Week 10	No-Code Prototyping: Make It Real	Applied no-code prototyping concepts to the PMTCT project and began developing the Healthy Start solution.	Deciding what to include in the first version.	Kept the prototype simple and focused on the core user need.	Prototype Link: Healthy Start: PMTCT Information & Access Guide 	
Week 11	AI Chatbots & Virtual Assistants	Explored how AI assistants and chatbots can support real tasks and considered where AI could add value to the PMTCT project.	Avoiding the use of AI simply because it was available.	Focused on whether AI genuinely improved the proposed solution.	AI Chatbots & Virtual Assistants session notes.	
Week 12	Test, Refine &	Tested the Healthy	Working with a	Treated the findings as		

	Measure Impact	Start prototype using a feedback form and reviewed the initial responses.	small number of responses.	initial feedback rather than generalising the results.		
Week 13	Communication & Pitching for Impact	Prepared my 90-second personal pitch and contributed content to the team pitch materials.	Presenting the project clearly without including unnecessary detail.	Focused on the problem, solution, evidence, testing and impact.		
Week 14	Project Report / Documentation	Updated the project documentation with the problem, evidence, blueprint, prototype, testing results, limitations and lessons learned.	Bringing the project evidence together in a clear structure.	Organised the evidence according to the project stages.		
Week 15	Final Assembly & Showcase Preparation	Reviewed my project materials, organised supporting evidence and prepared the project	Ensuring that the required evidence was documented before	Used a final checklist to identify and complete outstanding items.		

		for final presentation.	the deadline.			
Week 16	WALUAIS Showcase & Graduation	Prepared final project materials for submission and showcase.	Working within the final submission deadline.	Prioritised completion and documentation of required deliverables.		

Appendices

Appendix A – Attendance and Participation Records

Attendance was recorded on Zoom. The screenshots have been left out of this public version to protect other participants' privacy.

Item	Workshop/ Webinar Title	Date	Description / Caption	Proof / Attachment
A1	STEAM Problem Discovery & Systems Thinking	4 Aug 2026	Zoom attendance evidence showing participation in Session 1.	
A2	AI-Powered Data & Analytics	11 Aug 2026	Zoom attendance evidence showing participation in Session 2.	
A3	Ethics, AI Safety & Responsible Innovation	19 Aug 2026	Zoom attendance evidence showing participation in Session 3.	
A4	Project Blueprint	25 Aug 2026	Zoom attendance evidence showing participation in Session 4.	
A5	No-Code Prototyping: Make It Real	19 Sept 2026	Zoom attendance evidence showing participation in the session.	

A6	AI Tools Lab: Discover Your Stack	26 Sept 2026	Zoom attendance evidence showing participation in the session on exploring AI tools for project development	
A7	No-Code AI & Automation Workflows	30 Sept 2026	Evidence of engagement with the session, including my contribution in the YouTube discussion.	https://www.youtube.com/live/zS-vAs6CMs0?si=YnL2HxBDBfD15T9d
A8	AI Chatbots and Virtual Assistants	3 Oct 2026	Zoom attendance evidence showing participation in the session.	

Appendix B – Feedback Forms

Item	Workshop/ Webinar Title	Date	Description / Caption	Proof / Attachment
B1	STEAM Problem Discovery & Systems Thinking	4 Aug 2026	Key learning from the session included understanding the real problem before proposing solutions, separating assumptions from evidence, narrowing broad topics, and distinguishing a topic from a problem.	
B2	AI-Powered Data & Analytics	11 Aug 2026	Learning focused on turning assumptions into questions, starting with the questions to be answered before collecting data, choosing and verifying appropriate evidence, protecting personal information, and using data and AI responsibly.	
B3	Ethics, AI Safety & Responsible Innovation	19 Aug 2026	Learning focused on responsible AI, including bias, privacy, missing voices, misuse, safeguards, and human judgement. I also learned that some barriers may remain invisible in the data, that AI should support rather than replace human judgement, and that solutions should help people without causing harm.	
B4	From Problem to	25 Aug 2026	Learning focused on moving from problem to evidence, solution, prototype, testing, and	

	Project Blueprint		impact. I learned to let the evidence guide the solution, start with clarity before choosing tools, recognise that a problem does not always need AI, and understand the user journey before designing.	
B5	No-Code Prototyping: Make It Real	19 Sep 2026	I learned to start small and build when it makes sense. I also learned to think from the user's perspective, understand what the user actually needs, and define the user journey before choosing the technology.	
B6	AI Tools Lab: Discover Your Stack	26 Sep 2026	I learned that different AI tools can be used for different tasks and that clear prompts help produce better results. AI can support research, writing, analysis, data organisation and content creation. I also learned to choose a tool based on the problem being solved and to use AI to support, rather than replace, my thinking. I applied this by using different AI and automation tools for specific project tasks and comparing their usefulness. I learned to choose tools based on the task and to verify AI-generated outputs rather than relying on them automatically.	
B7	AI Automation &	30 Sep 2026	I learned that automation can make repeated tasks happen automatically and that a workflow can move information from one step to another	

	Workflows in Practice		without doing everything manually. I practised simple notification automation for the team document. This was a learning exercise and was not part of the team project. I have not yet applied automation to the team project.	
B8	AI Chatbots & Virtual Assistants	3 Oct 2026	I learned how AI chatbots and virtual assistants can support real tasks and that they should only be used when they add value. I also learned to consider AI capabilities, limitations, and responsible use before choosing an AI-based solution. I applied this learning by considering whether AI assistance was necessary for the PMTCT project and keeping the Healthy Start solution simple and focused on the user need.	Women Ascension …

Appendix C – Raw Data & Notes

Item	Workshop/ Webinar Title	Date	Description / Caption	Proof / Attachment
C1	AI & STEAM Problem Discovery & Systems Thinking	4 Aug 2026	Problem discovery working material, including the PMTCT problem statement, systems thinking analysis, and investigation questions.	
C2	AI-Powered Data & Analytics	11 Aug 2026	Individual Assignment 2: Data Thinking & Evidence, including the transportation question, Google Form, sample responses, analysis, insights, and visual.	
C3	Ethics, AI Safety & Responsible Innovation	19 Aug 2026	Responsible AI session notes covering bias, privacy, missing voices, safeguards, human judgement, and responsible use of AI.	Responsible AI session notes and 10 Rule Ethical AI Checklist.
C4	Healthy Start: PMTCT Information & Access Guide	Sep–Oct 2026	Team project working materials, including the Healthy Start prototype, user feedback form, testing results, project documentation, and pitch materials.	https://obiageli-e.github.io/healthy-start-pmtct/